

Course: Power Transmission Networks

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Project II

Topic 7: Mathematical Modelling of a HV System as a Total & Specification of System Operation Modes

1. Introduction

Mathematical modelling of a High Voltage (HV) system "as a total" generally refers to **Power Flow (or Load Flow) Analysis**. This is the fundamental tool used to determine the steady-state operating condition of a power system. Unlike looking at a single line or generator in isolation, this model connects all components—generators, transformers, transmission lines, and loads—into a unified mathematical system.

The primary goal of this modelling is to calculate the voltage magnitude ($|V|$) and voltage angle (δ) at every bus (node) in the system. Once these are known, the active power (P) and reactive power (Q) flows through all transmission lines can be calculated, ensuring the system operates within stable limits.

2. Mathematical Formulation of the Network

To model the HV system mathematically, we treat the grid as a network of electrical nodes (buses) connected by branches (lines/transformers). The most efficient way to represent this connectivity for computer calculation is using the **Bus Admittance Matrix (Y_{bus})**.

2.1 The Y-Bus Matrix

The system is modelled using Nodal Analysis based on Kirchhoff's Current Law (KCL). For an n -bus system, the equation is:

$$[I] = [Y_{bus}] * [V]$$

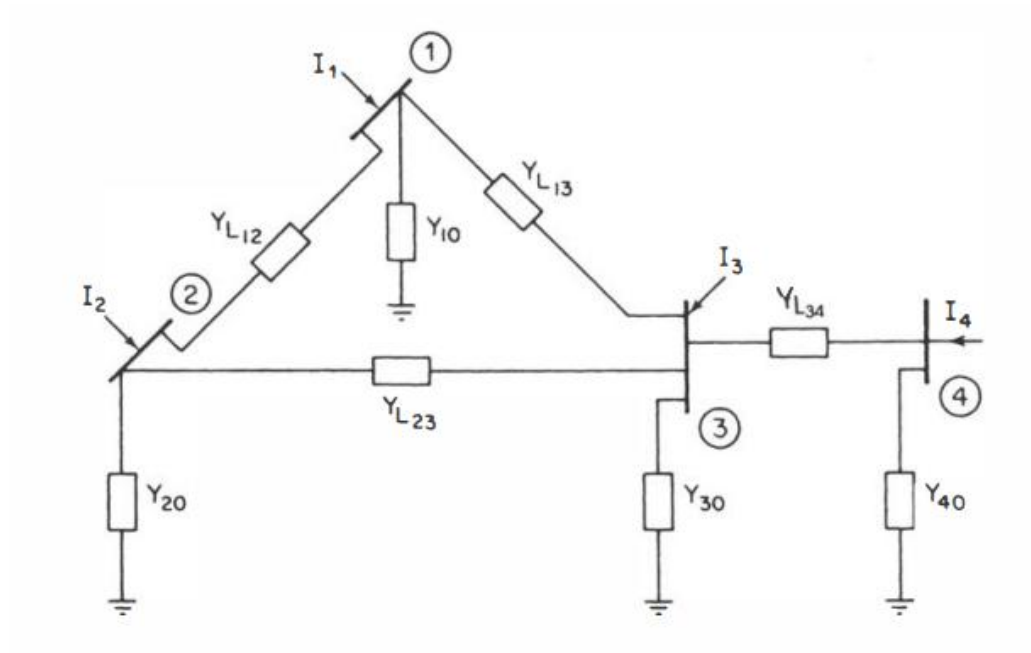


Figure 1 Equivalent Circuit for System

Where:

- $[I]$ is the vector of injected currents at the buses.
- $[V]$ is the vector of complex bus voltages.
- $[Y_{bus}]$ is the $n \times n$ admittance matrix.

The elements of the Y_{bus} matrix are constructed as follows:

- **Diagonal elements (Y_{ii}):** The sum of all admittances connected to bus i (including shunts).
- **Off-diagonal elements (Y_{ij}):** The negative of the admittance connecting bus i and bus j .

2.2 The Power Flow Equations

Since power systems operate at constant voltage rather than constant current, we replace current I with power S . The complex power injected at bus i is given by:

$$S_i = P_i + jQ_i = V_i \cdot I_i^*$$

Substituting the Y_{bus} relationship into this power equation results in the **Static Load Flow Equations**. For a specific bus i , the net real (P_i) and reactive (Q_i) power injected is:

$$P_i = \sum_{j=1}^n |V_i||V_j||Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j)$$

$$Q_i = - \sum_{j=1}^n |V_i||V_j||Y_{ij}| \sin(\theta_{ij} - \delta_i + \delta_j)$$

3. Specification of System Operation Modes (Bus Classification)

A critical part of modelling the HV system is defining the "Operation Modes" of the various nodes. In a mathematical sense, this is done by classifying every bus in the system into one of three types based on which variables are **known (specified)** and which are **unknown (calculated)**.

There are four key variables at each bus:

1. Active Power (P)
2. Reactive Power (Q)
3. Voltage Magnitude ($|V|$)
4. Voltage Angle (δ)

Because we have two equations per bus (one for P , one for Q), we must specify two variables and solve for the other two.

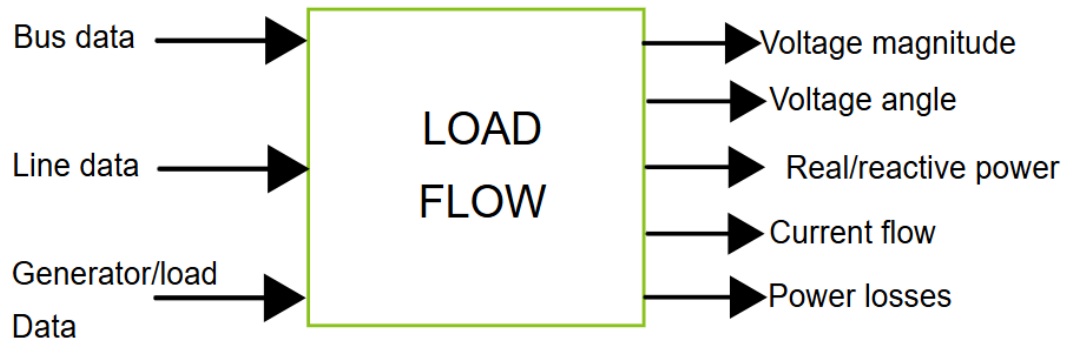


Figure 2 The inputs and outputs of load flow analysis

3.1 PV Bus (Generator Bus / Voltage Controlled Bus)

- **Physical equivalent:** A bus connected to a generator or a synchronous condenser where the voltage can be regulated.
- **Specified (Known):** Active Power (P_G) and Voltage Magnitude ($[V]$).
- **Unknown (To be calculated):** Reactive Power (Q_G) and Voltage Angle (δ).
- **Constraint:** The calculated Q must remain within the physical limits of the generator ($Q_{min} \leq Q \leq Q_{max}$). If the limit is violated, this bus may be switched to a PQ bus type.

3.2 PQ Bus (Load Bus)

- **Physical equivalent:** Substations serving consumers or buses where no generation is present. This is the most common bus type (approx. 80-90% of buses in a model).
- **Specified (Known):** Active Power (P_L) and Reactive Power (Q_L).
Note: $P_{inj} = -P_{load}$.
- **Unknown (To be calculated):** Voltage Magnitude ($[V]$) and Voltage Angle (δ).

3.3 Slack Bus (Swing Bus / Reference Bus)

- **Physical equivalent:** Usually the largest generator in the system (e.g., a hydro dam or nuclear plant).

- **Role:** In any power system, the total generation must exactly equal total load plus total losses (P_{loss}). However, $P_{loss} (I^2 * R)$ cannot be calculated until the solution is finished (since currents are unknown). Therefore, one generator must be left "free" to supply whatever remaining power is needed to balance the system.
- **Specified (Known):** Voltage Magnitude ($|V| \approx 1.0$ p.u) and Voltage Angle ($\delta = 0$).
- **Unknown (To be calculated):** Active Power (P) and Reactive Power (Q).

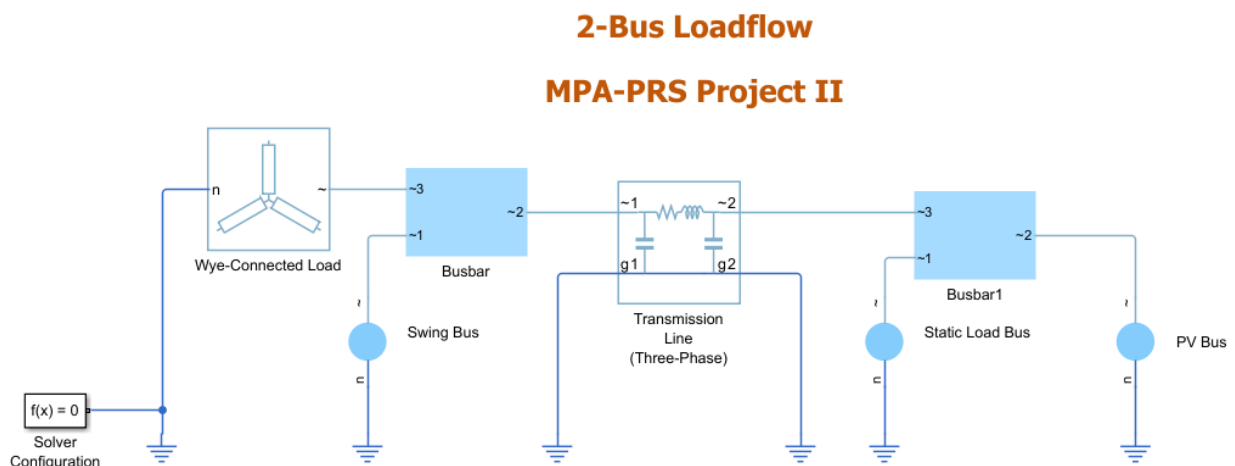


Figure 3 Simple Load Flow Analysis on Simulink Simspace

Conclusion

Mathematical modelling of the HV system as a total requires the construction of the Admittance Matrix and the formulation of non-linear power flow equations. By specifying the operation modes of system buses (**Slack**, **PV**, and **PQ**), engineers can solve these equations to determine the voltage profile and power flows, ensuring the grid is capable of meeting demand safely and efficiently.

List of References

1. El-Hawary, M. E. (2008). *Electrical Power Systems: Design and Analysis*. IEEE Press. (Specifically, pp. 300-311) Retrieved from <https://cjitsee.files.wordpress.com/2012/12/electrical-power-systems-el-hawary.pdf>
2. Electrism. (n.d.). *Load Flow / Power Flow Analysis*. Retrieved from <https://electrism.com/load-flow-power-flow.html>

Topic 9: Reactors, Capacitors, and Synchronous Compensators on HV System

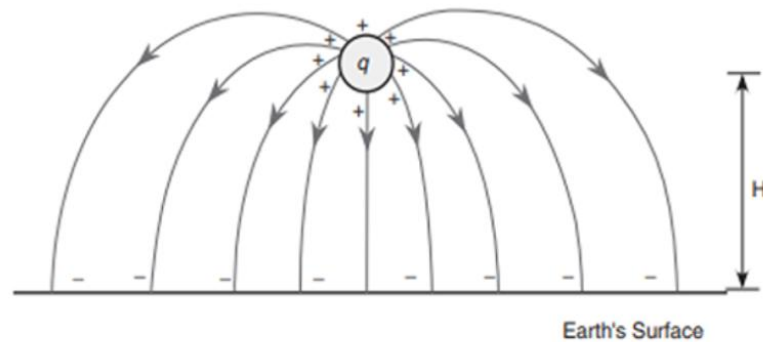
1. Introduction to Reactive Power Management

In High Voltage (HV) transmission systems, the management of **Reactive Power (Q)** is a fundamental requirement for maintaining voltage stability. While Active Power (**P**) represents the useful energy transferred to loads (lighting, mechanical work), Reactive Power represents the oscillating energy stored in the electric and magnetic fields of the system components. Transmission lines act simultaneously as capacitors (due to the potential difference between conductors) and inductors (due to current flow creating magnetic fields). To maintain the voltage profile within the required regulatory limits (typically 5%), utilities must deploy **Compensation Equipment**. This equipment is broadly categorized into static devices (Reactors and Capacitors) and dynamic machines (Synchronous Condensers).

2. Shunt Reactors

A Shunt Reactor is a high-inductance coil connected between a phase conductor and the ground. Its primary purpose is to absorb reactive power (**Q**). Construction wise, these can be "dry-type" (air-core) for lower voltages or "oil-immersed" (iron core) for EHV applications to provide insulation and cooling. The reactive power absorbed is defined by:

$$Q_{absorbed} = \frac{|V|^2}{X_L} = \frac{|V|^2}{2\pi f L}$$



Distribution of electric field lines from an overhead conductor to earth's surface.

Figure 4 Electric field distribution and capacitance in transmission lines.

Addressing the Ferranti Effect

The critical application of shunt reactors is mitigating the **Ferranti Effect**.

- **Phenomenon:** On long, lightly loaded transmission lines (or open-ended lines), the line's inherent shunt capacitance generates significant reactive power (Q_c). As these capacitive current flows through the line inductance towards the source, it causes a voltage rise. Consequently, the receiving end voltage (V_R) becomes significantly higher than the sending end voltage (V_S).
- **Impact:** This overvoltage can stress insulation, trigger lightning arresters, or damage substation equipment.
- **Operation:** Shunt reactors are typically installed at the ends of long EHV lines. By absorbing the excess charging current (I_c), they neutralize the capacitive effect, keeping VR within safe limits.

Switching Transients

Shunt reactors are often switched on and off daily. This frequent switching can cause "current chopping," leading to high transient recovery voltages (TRV). Modern circuit breakers for reactors are equipped with "opening resistors" or synchronized switching controllers to mitigate these transients.

3. Shunt Capacitors

Mechanism of Voltage Support: Shunt Capacitor Banks are the most economical method for supplying reactive power. They consist of series-parallel combinations of capacitor cans.

During heavy load periods, the system voltage tends to sag due to:

1. The inductive nature of most loads (motors).
2. The I^2X_L voltage drop across the transmission line itself.

By connecting capacitors in shunt, we inject reactive power (Q_c) locally at the load bus. This reduces the reactive current component that must be transmitted from the generator, reducing the overall current magnitude (I).

$$Q_{\text{injected}} = |V|^2 \cdot \omega C$$

Benefits and Limitations

- **Benefit-Power Factor Correction:** By bringing the load power factor closer to unity, line losses ($P_{loss} = I^2 * R$) are significantly reduced, releasing thermal capacity in transformers and cables.
- **Limitation- Voltage Dependency:** A major drawback of static capacitors is that their output is proportional to the square of the voltage (V^2). During a grid fault, when voltage drops significantly, the capacitor's ability to support the system drops drastically—exactly when support is needed most.

4. Synchronous Compensators (Condensers)

Operational Physics

A Synchronous Condenser is a synchronous motor running without a mechanical load. It floats on the grid. Its reactive behaviour is determined by the DC excitation current (I_f) applied to the rotor field winding, described by the "V-Curve" of the machine.

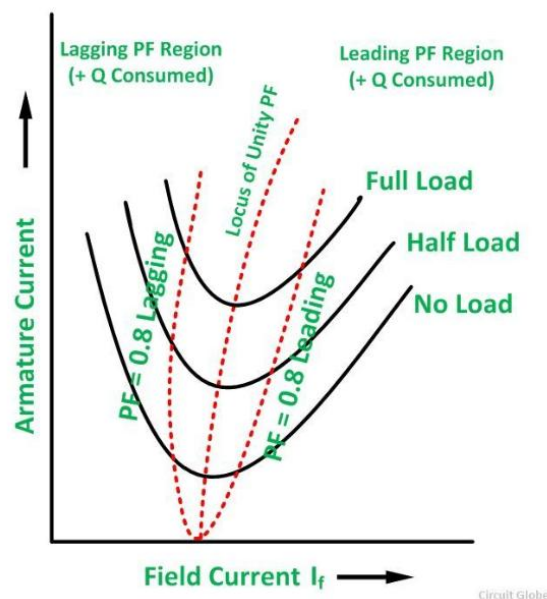


Figure 5 V-Curve of a Synchronous Machine showing lagging and leading

- **Over-Excited Mode:** If excitation is high ($E > V$), the machine pushes current into the grid, acting as a variable capacitor (Source of Q).
- **Under-Excited Mode:** If excitation is low ($E < V$), the machine draws magnetizing current from the grid, acting as a variable reactor (Sink of Q).

Unlike switched capacitors/reactors which provide step-changes in voltage, a synchronous condenser provides smooth, stepless control. Furthermore, because it is a rotating machine with significant mass, it provides System Inertia.

Conclusion

The selection between these devices involves a trade-off. Capacitors and reactors are cheap and simple but static. Synchronous condensers are expensive and require maintenance but offer superior dynamic performance and inertia. Modern HV systems typically use a hybrid approach, or increasingly, power-electronics based STATCOMs which mimic synchronous condensers without the moving parts.

References

1. Saadat, H.(2010). Power System Analysis. (Chapter: Transmission Line Parameters, pp. 165-171). Retrieved from <http://powerunit-ju.com/wp-content/uploads/2016/11/Power-System-Analysis-by-HadiSaadat-Electrical-Engineering-libre.pdf>
2. Slideshare (Khemraj). Series and Shunt Compensation.

Topic 10: Transmission capacity and parameters compensation of power lines. Compensation equipment's positioning.

Determinants of Transmission Capacity

The capacity of a power line is not a single fixed number. It is a dynamic limit determined by the most restrictive of three physical constraints depending on the line's length.

A. Thermal Limit (Short Lines < 80 km)

The physical conductors (typically ACSR- Aluminium Conductor Steel Reinforced) have a maximum operating temperature (e.g., 100 degree C). Exceeding this causes:

- **Sagging:** The metal expands, lowering the line clearance to ground, creating a safety hazard (flashover risk).
- **Annealing:** Permanent loss of mechanical tensile strength. Thermal capacity varies with ambient conditions; a line can carry more current on a cold, windy day than on a hot, still day. This has led to Dynamic Line Rating (DLR) technologies.

B. Voltage Drop Limit (Medium Lines 80-320 km)

For medium lengths, the impedance causes a voltage drop ($IR + jIX$). The system operator is mandated to keep the receiving end voltage (V_R) within 5% of nominal. Even if the wire is not melting (thermal limit safe), the capacity is limited if the voltage at the consumer end drops too low.

C. Stability Limit (Long Lines > 320 km)

For very long lines, the dominant constraint is Steady-State Stability. The power transferred between two buses with voltage angles δ_s and δ_r is:

$$P = \frac{|V_S||V_R|}{X_{total}} \sin(\delta)$$

Theoretical max power occurs at $\delta = 90$ deg. However, for safety margin against transient disturbances, operation is usually limited to 30 – 40 deg. This limit is often far below the thermal capacity of the conductor.

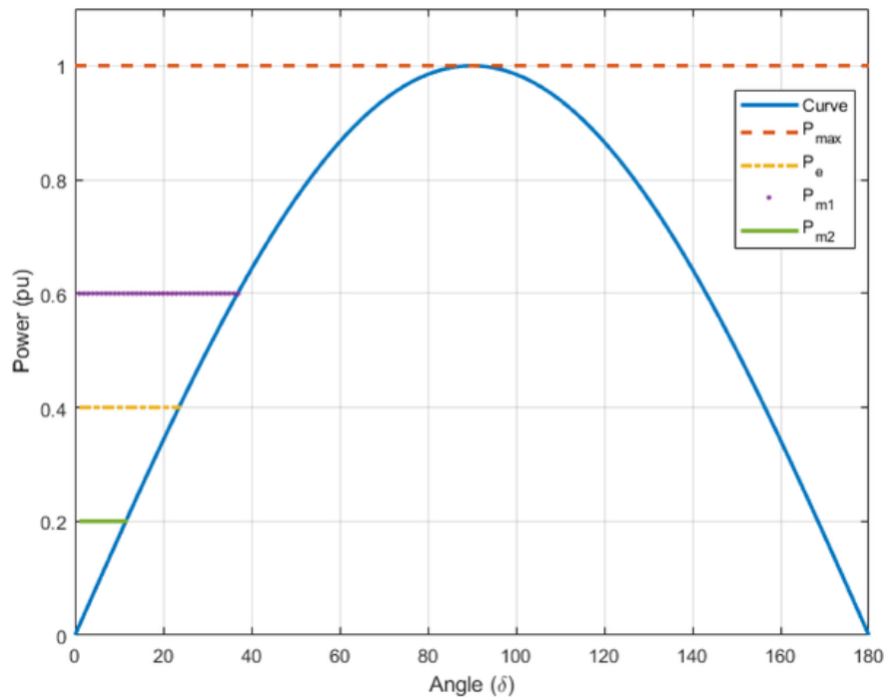


Figure 6 Power transfer capability vs. Load Angle (δ).

Surge Impedance Loading (SIL)

The natural loading of a line is its Surge Impedance Loading.

$$SIL = \frac{V_{LL}^2}{Z_c} \quad \text{where} \quad Z_c = \sqrt{\frac{L}{C}}$$

- When Power Flow $P = SIL$: The reactive power generated by line capacitance exactly equals the active power absorbed by line inductance. Voltage profile is flat.
- When $P > SIL$: The line is net inductive (voltage drops).
- When $P < SIL$: The line is net capacitive (voltage rises).

For 400kV lines, **SIL** is typically around 500-600 MW. Loading beyond this requires significant compensation.

Series Compensation and Positioning

To overcome the Stability Limit (Constraint C), we must maximize the power equation. Since we cannot easily change V or δ , we reduce X . Series Compensation involves inserting capacitors in series with the line.

$$X_{new} = X_{line} - X_{capacitor}$$

If we compensate 50% of the reactance ($k = 0.5$), we effectively double the electrical shortness of the line, doubling its stability limit.

Positioning Strategies

The physical location of compensation equipment significantly impacts the volt age profile along the line.

1. Mid-Point Compensation: Placing a shunt capacitor or series capacitor at the exact electrical centre of a long line is theoretically optimal. It effectively segments the line into two halves. This segmentation drastically increases the stability limit compared to placing the same equipment at the ends.

2. End-Point Compensation: Typically found in substations. While less effective for increasing stability limits than mid-point compensation, it is far cheaper and easier to maintain.

3. Distributed Compensation: Modern Flexible AC Transmission Systems (FACTS) like TCSC (Thyristor Controlled Series Capacitors) allow for variable compensation, effectively changing the position of compensation electrically to suit real-time grid conditions.

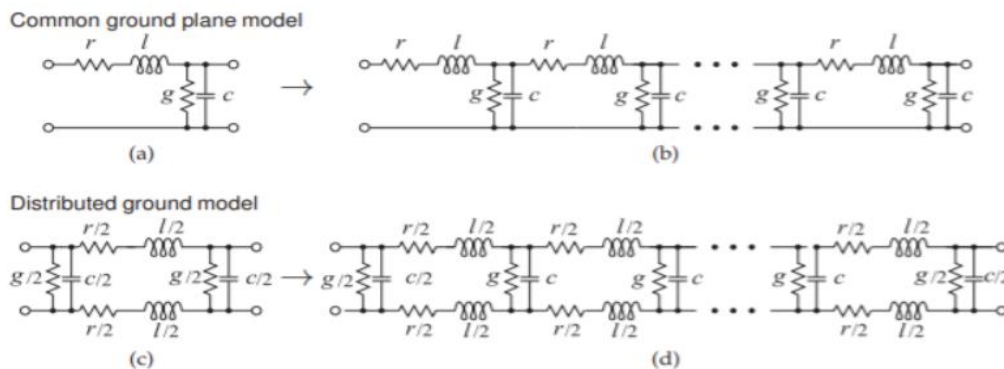


Figure 7 Distributed parameter model of a long transmission line.

Risks: Sub-Synchronous Resonance (SSR)

A critical design consideration for series compensation is SSR. This occurs when the electrical resonance frequency of the series capacitor and line inductance interacts with the mechanical torsional frequency of the turbine-generator shaft. This resonance can physically snap turbine shafts, requiring specialized damping equipment.

References

1. Von Meier, A. (2006). Electric Power Systems: A Conceptual Introduction. (pp. 182-184). Retrieved from <https://www.personal.psu.edu/sab51/vls/vonmeier.pdf>
2. IJET. (2016). Transmission Line Compensation
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Topic 14: Short-Circuit Calculations

The Necessity of Fault Analysis

Short-circuit analysis is the most critical safety calculation in power system design. Unlike load flow (which looks at normal operation), fault analysis looks at the worst-case scenario: the breakdown of insulation. The results of these calculations determine:

- **Circuit Breaker Ratings:** The maximum current a breaker must interrupt.
- **Relay Settings:** The thresholds at which protection devices trigger.
- **Busbar Mechanical Strength:** Fault currents create massive magnetic forces that can bend copper busbars.

Classification of Faults

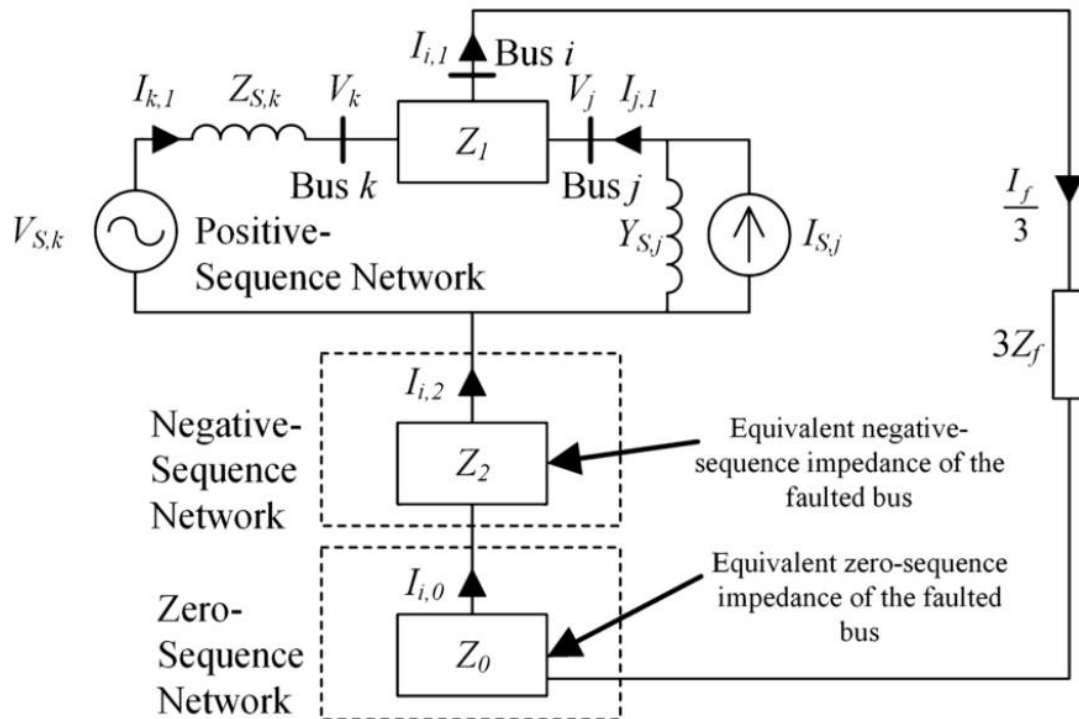
- **3-Phase Fault (L-L-L):** All three phases shorted together. This is the simplest to calculate but usually yields the highest current. It is the benchmark for equipment sizing.
- **Single Line-to-Ground (SLG):** The most common fault (> 70%), usually caused by lightning, trees, or insulator failure.
- **Line-to-Line (L-L):** Two phases touching, often due to wind-induced galloping.
- **Double Line-to-Ground (DLG):** Two phases grounded.

Calculation Methodology: The Per-Unit System

To perform these calculations across different voltage levels (from 400kV transmission down to 11kV distribution), engineers use the Per-Unit (p.u.) System.

$$Z_{p.u.} = Z_{actual}(\Omega) \times \frac{S_{base}}{V_{base}^2}$$

This normalizes all impedances, allowing the entire grid to be solved as a single circuit without complex transformer ratio calculations.



Sequence networks during a single line-to-ground fault

Figure 8 Interconnection of Sequence Networks for L-G Fault

The Three Sequences

- **Positive Sequence (+):** Represents normal system operation (source rotation).
- **Negative Sequence (-):** Represents reverse rotation. Arises only during unbalance.
- **Zero Sequence (0):** Represents current returning through the ground/neutral.

Network Interconnection for Faults

The type of fault dictates how these three abstract networks are connected mathematically:

- **L-G Fault:** The three networks (+ 0) are connected in Series.
- **L-L Fault:** The Positive and Negative networks are connected in Parallel (Zero sequence is not involved).
- **L-L-L Fault:** Only the Positive sequence network is used (Negative and Zero are zero).

Breaker Ratings: Making vs. Breaking Capacity

A critical distinction in short-circuit calculations is the timing of the current.

Breaking Capacity (AC Component)

This is the RMS value of the current at the moment the breaker contacts separate (usually 3 to 5 cycles after the fault begins). This determines if the breaker can extinguish the arc.

Making Capacity (Peak DC Offset)

Fault currents are not perfect sine waves immediately. They contain a decaying **DC** Offset. The first peak of the fault current is the highest (asymmetrical peak) and can be 2.5 times the RMS value. The Making Capacity is the ability of a breaker to close onto a fault safely. If a breaker closes on a fault, it must withstand the immense magnetic forces of this initial peak current without exploding.

$$I_{peak} \approx \sqrt{2} \cdot I_{rms} \cdot (1 + e^{-t/\tau})$$

Conclusion

Short-circuit calculations involve reducing the complex grid to a Thevenin Equivalent impedance (Z_{th}) at the fault point. By utilizing Symmetrical Components, engineers can calculate the currents for all fault types, ensuring that protection systems are robust enough to isolate faults before permanent equipment damage occurs.

References

1. El-Hawary, M. E. (2008). Electrical Power Systems. (Chapter: Fault Analysis, pp. 469-518). Retrieved from <https://cjitseee.files.wordpress.com/2012/12/electrical-power-systems-el-hawary.pdf>